ALPANA_SOC

Our own processor chip design, built on the open RISC-V standard and running on an FPGA board

80 MHz	1.5×	28×	Open
clock speed	faster than the original Shakti design	faster at decimal (floating-point) maths	no licence fees, full design ownership

What is it?

ALPANA_SOC is a complete small computer on a single chip (a "System-on-Chip"): a processor, memory controller and connections to sensors and motors. It is designed by us and runs on an FPGA, a reprogrammable chip that lets us test hardware without manufacturing silicon. It is based on **Shakti**, India's open-source processor from IIT Madras, which we have upgraded.

What we improved

- **Faster clock (50 → 80 MHz).** We split the processor's work into 4 smaller steps instead of 3, like adding one more worker to an assembly line. Each step is simpler, so the whole chip runs faster.
- **Built-in maths engine (FPU).** The original chip had to do decimal maths (like 3.14 × 2.5) slowly in software. Ours does it in hardware: **28× faster**. That matters for robots and drones, which constantly calculate angles, speeds and positions.
- **Tested thoroughly.** It passes the industry-standard RISC-V test suite and our own stress tests, and we found and fixed two hidden bugs in the original design.
- **Ready to connect hardware.** It has ports for motors (PWM), sensors (I^2C/SPI), GPS and serial links (UART), Ethernet, and 256 MB of memory.

How it compares

	ALPANA_SOC	Original Shakti	Typical microcontroller (Arm)
Speed (clock)	80 MHz	50 MHz	Fixed by the manufacturer
Decimal maths	Hardware, fast	Software, slow	Only on some models
Can we change the design?	Yes, fully	Yes	No: closed design
Licence cost	None	None	Paid
Add custom hardware	Yes	Yes	No

Why it matters: owning the processor design means we can add exactly the features an application needs (a camera interface, a motor controller, a security block) instead of being limited to what an off-the-shelf chip offers.

Performance in numbers

Test	Original Shakti	ALPANA_SOC	Better by
Dhrystone (general computing score)	5.7	8.6	1.5×
CoreMark (industry benchmark)	21	32	1.5×
Decimal-maths task (matrix multiply)	112 ms	4 ms	28×

Higher is better for the scores; lower is better for time. Measured in simulation of the actual chip design; board measurements to follow.

Applications

Application	What the chip does
Autonomous buggy	Drives the motors, reads distance sensors and steers around obstacles on its own.
Camera interfacing	Connects to a small camera, captures images and does simple processing (e.g. detecting colours or edges).
Drone controller	Reads motion sensors, calculates the drone's tilt many times a second, and adjusts the motor speeds to keep it stable. The fast maths engine is key here.
Other uses	Smart sensors, motor control, education and research, security hardware.

Live demonstration

Available from 21 October 2026 (one week from now), on the FPGA board.

#	What you will see
1	The chip starting up and running benchmarks live, with results on screen
2	Autonomous buggy: motors controlled by ALPANA, avoiding obstacles
3	Camera: images captured and processed by ALPANA
4	Drone controller: sensor reading and motor control on the bench (propellers off)

Status: chip design complete and tested in simulation. This week: final board testing and the demo applications.